

ORIGINAL

Sustainable tourism and environmental impact assessment based on Big Data

Turismo sostenible y evaluación de impacto ambiental basada en Big Data

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ABSTRACT

Introduction: big data enables monitoring of tourism's environmental impacts through mobility, satellite, and social data. Existing research applies machine learning for prediction and assessment. However, existing research face limitations in data integration, scalability, privacy concerns, and a lack of standardized frameworks for sustainable tourism decision-making.

Objective: this research proposed a big data-driven analysis for sustainable tourism and environmental impact assessment using advanced machine learning techniques.

Method: the Kaggle platform provides access to a dataset of 9,321 samples on tourism and environmental impact, collected from relevant tourism authorities and environmental agencies. The data involve rigorous preprocessing, including data cleaning with missing value handling, which prevents bias and incomplete records, and Min-Max normalization to scales standardizes, ensuring quality and consistency. Feature extraction is followed by Linear Discriminant Analysis (LDA) to identify hidden patterns and correlations between tourism intensity and environmental degradation indicators such as land-use change, vegetation density, and resource consumption. The Killer Whale Algorithm-tuned Weighted Random Forest (KWA-Weighted RF) is used in predictive modelling to estimate tourism demand and assess environmental effect trends. KWA optimizes hyperparameters, improving both search efficiency and prediction accuracy. Weighted RF increases robustness by emphasizing critical characteristics and managing imbalance.

Results: experimental results demonstrate that the proposed Python-based big data-driven models achieve accuracy (98,7 %), precision (96,3 %), recall (95,1 %), F1-score (95,7 %), Kappa (0,964), MAE (0,112), RMSE (0,168), and AUC (0,982) in both tourism demand estimation and environmental impact classification.

Conclusions: the findings highlight that sustainable tourism management can be significantly enhanced through data analytics and AI-based decision support systems.

Keywords: Sustainable Tourism; Environmental Impact Assessment (EIA); Big Data Analytics; Machine Learning; Ecological Sustainability; Tourism Demand Forecasting; Geospatial Analysis.

RESUMEN

Introducción: big Data permite monitorear los impactos ambientales del turismo mediante datos de movilidad, satelitales y sociales. Las investigaciones existentes aplican aprendizaje automático para la predicción y la evaluación. Sin embargo, dichas investigaciones enfrentan limitaciones en la integración de datos, la escalabilidad, las preocupaciones sobre la privacidad y la falta de marcos estandarizados para la toma de decisiones en turismo sostenible.

Objetivo: esta investigación propone un análisis impulsado por Big Data para la evaluación del turismo sostenible y de los impactos ambientales utilizando técnicas avanzadas de aprendizaje automático.

Método: la plataforma Kaggle proporciona acceso a un conjunto de datos de 9,321 muestras sobre turismo e impacto ambiental, recogidas de autoridades turísticas y agencias ambientales relevantes. Los datos pasan por un riguroso preprocesamiento, que incluye limpieza de datos con manejo de valores faltantes para evitar sesgos y registros incompletos, y normalización Min-Max para escalar y estandarizar los datos, garantizando calidad y consistencia. Tras la extracción de características se aplica un Análisis Discriminante Lineal (LDA) para identificar patrones ocultos y correlaciones entre la intensidad turística y los indicadores de degradación ambiental, como el cambio en el uso del suelo, la densidad de vegetación y el consumo de recursos. Para la modelización predictiva se emplea un Random Forest ponderado ajustado mediante el Algoritmo Killer Whale (KWA-Weighted RF) para estimar la demanda turística y evaluar las tendencias de los efectos ambientales. KWA optimiza los hiperparámetros, mejorando tanto la eficiencia de búsqueda como la precisión de la predicción. El Random Forest ponderado incrementa la robustez al enfatizar las características críticas y gestionar el desequilibrio de clases.

Resultados: los resultados experimentales demuestran que los modelos propuestos, basados en Python e impulsados por Big Data, alcanzan exactitud (98,7 %), precisión (96,3 %), sensibilidad (recall) (95,1 %), puntuación F1 (95,7 %), Kappa (0,964), MAE (0,112), RMSE (0,168) y AUC (0,982) tanto en la estimación de la demanda turística como en la clasificación del impacto ambiental.

Conclusiones: los hallazgos destacan que la gestión del turismo sostenible puede mejorarse significativamente mediante el análisis de datos y sistemas de apoyo a la toma de decisiones basados en IA.

Palabras clave: Turismo Sostenible; Evaluación de Impacto Ambiental (EIA); Análisis de Big Data; Aprendizaje Automático; Sostenibilidad Ecológica; Pronóstico de Demanda Turística; Análisis Geoespacial.

INTRODUCTION

Tourism leads to increased job creation and cultural interaction, it poses several environmental problems, such as pollution, biodiversity depletion, garbage disposal, and energy and water usage.⁽¹⁾ Sustainable tourism involves creating a balance between development, protection, and social wellbeing while ensuring minimal degradation, maintaining culture, and supporting the sustainable development of regions around the world.

⁽²⁾ An Environmental Impact Assessment (EIA) assesses potential impacts to be expected from tourists before implementation and minimizes any negative effects by considering available data, which has been limited and does not account for dynamic changes.⁽³⁾

Big data in tourism is used in enhancing EIA by analyzing huge amounts of electronic data to provide accurate and quick analysis.⁽⁴⁾ Big data refers to high-dimensional, multi-source structured and unstructured tourism and environmental datasets generated from heterogeneous platforms, including social media, GPS mobility data, satellite imagery, mobile applications, and web-based systems.⁽⁵⁾ Deep learning models analyze tourist visitation patterns to optimize flow, reduce congestion, and support sustainable resource use.⁽⁶⁾ Intelligent tourism solutions based on big data detect anomalies, enhance safety, optimize operations, and allow real-time management decisions based on data.⁽⁷⁾ Making use of large datasets, ensemble deep learning algorithms predict the demand for tourism on a daily basis, improving the accuracy of forecasting, planning, and controlling the efficiency of crowds.⁽⁸⁾

The literature reviews diverse strategies that demonstrated the need to have more integrated sustainable tourism strategies. A K-nearest neighbor (KNN) data-driven technique produced personalized Nepal travel suggestions, which were assessed using an F1-score (83,4 %) and AUC (94,5 %). However, scalability remained restricted, and performance was strongly reliant on the quality and dependability of previously given user data.

⁽⁹⁾ A long short-term memory network (LSTM)-based method was used for long-term tourism demand forecasting. The assessment findings show that the suggested models outperform MAPE (4.533 % for Hotel A, 16.986 % for Hotel B, and 8.336 % for Hotel C). Large datasets and enough computer power were necessary.⁽¹⁰⁾ An attention mechanism bidirectional long short-term memory (ATT-BiLSTM) forecasting method was proposed to that enhances data feature extraction. The results show that the proposed model not only has better generalizability, but it also outperforms the RMSE by 3,26 % before and 7,71 % after COVID-19; data heterogeneity complicates preprocessing.⁽¹¹⁾

Research using a variety of geographic data sources assessed the viability of ecotourism in Zhangjiajie, China. The forecast results of Suitability were as follows: extremely appropriate (19,34 %), moderate (28,78 %), marginal (23,87 %), and unsuitable (28,01 %). The percentage of the area, including the case-specific characteristics, hindrance to generalization.⁽¹²⁾ The convergence of sustainable and green tourism with the Sustainable Development Goals (SDG) is focused on the importance of tourism in the balance between economic growth, environmental maintenance, and social justice, and the importance of high-quality statistics to interpret it correctly and make a decision.⁽¹³⁾

Smarter Sustainable Tourism applies big data deep-learning to analyze multi-level tourism insights and

derive metrics based on Twitter, yet it needs concrete real-world validation to apply in a practical smart tourism system.⁽¹⁴⁾ Tourism economic effects were analyzed in Saudi Arabia during the COVID-19 pandemic as a basis for making future policies, yet there is a lack of information from different regions, preventing any conclusions on a global level.⁽¹⁵⁾ Documentation Framework Architectural Decisions (DFAD) approach integrates big data management into digital tourism models and contributes to better visitor behavior and spending, yet it is theoretical and lacks any practical applications and research validation.⁽¹⁶⁾ Application of artificial intelligence in geographic data visualization and analysis in terms of low-carbon rural slow tourism. The study found out that K-Means cluster shows 5,26 %, while support vector machine - 19,2 % due to the vector's characteristics.⁽¹⁷⁾

An ontology-based semantic modelling context-aware recommendation system through deep neural networks that sustains tourism through experience and ecological conservation despite challenges in building an advanced ontology.⁽¹⁸⁾ An integration of assessment models for ecotourism capability analysis in Sundarban Biosphere Reserve of India using Deep Learning Neural Networks (DLNN). Consequently, the DLNN model's AUC is 0,91, and the weight of parameter-dependent models.⁽¹⁹⁾ A data mining algorithm using Support Vector Machine and Red Deer Optimization (SVM-RDO). The proposed SVM-RDO has surpassed previous algorithms for tourist datasets, delivering an accuracy rate of 92 % and F1-score of 89,8 % through parameter changes.⁽²⁰⁾

Research gap: however, most of the existing research have been focused on single task prediction with the use of local/regional databases. Very few studies have taken into account multi-source big data analysis and sustainability indices.^(10,16) Scalability, generalizability, and real-time decision making are the reasons for the need for a consistent and interpretable model for comprehensive evaluation of sustainable tourism development and impact.^(19,20) The proposed research addresses these limitations by developing a big data framework based on a KWA-weighted Random Forest model to improve scalability, generalization, and interpretability for forecasting tourism demand and assessing environmental impact using sustainability indices, integrated with LDA-based feature learning for robust sustainability prediction.

Key Contributions

- Integration of multi-source tourism and environmental datasets using advanced pre-processing for improved data quality and consistency.
- Application of LDA-based feature extraction to uncover hidden relationships between tourism activity and environmental indicators.
- Development of a KWA-tuned Weighted Random Forest model for accurate, scalable, and interpretable prediction of tourism demand and environmental impact.

METHOD

The big data tourism-environment dataset is processed through data cleaning, min-max normalized, LDA feature extraction, KWA optimization, and weighted RF prediction, and its performance is measured using metrics. Figure 1 depicts the tourism demand and environmental effect model.

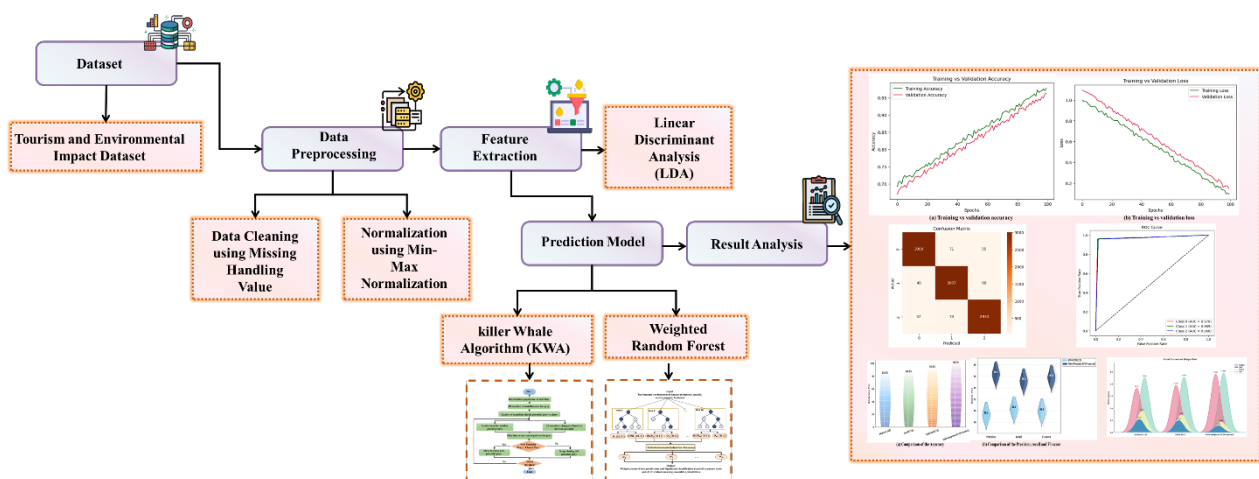


Figure 1. Overall Tourism Prediction Model

Dataset

A dataset on tourism and environmental effects is obtained via the Kaggle platform (<https://www.kaggle.com/datasets/programmer3/tourism-and-environmental-impact-dataset/data>), and is distributed under the CC0 (Public Domain) license, allowing unrestricted reuse, modification without copyright restrictions. The dataset consists of 9,321 samples in one CSV file, including tourist intensity, environmental indicators, and

socioeconomic aspects such as visitor flow, seasonality, mobility, lodging, transportation demand, land-use, vegetation, air quality, water, waste, and energy.

Data preprocessing

Preprocessing of data involves cleaning of data, handling of missing data, and Minmax normalization to ensure consistency, precision, and improved model performance.

Data Cleaning

The data cleaning phase removed errors, missing values, noise, and duplicates. Missing data was handled using interpolation and mean imputation techniques, whereas noise reduction was performed to improve signal quality, producing consistent and accurate data reflecting tourism activity and environment variables.

Min-Max Normalization

Min-Max normalization is applied, and class imbalance is handled during preprocessing to ensure fair learning across classes by remapping features to a fixed range, making the data spread uniformly, enhancing model stability and comparability, and avoiding bias due to the variations in feature magnitudes, which ensures that the features in machine learning models are.

$$M_{norm} = \frac{M - \min(M)}{\max(M) - \min(M)} \quad (1)$$

Here, $\max(M)$ refers to the greatest value of property, and $\min(M)$ is the smallest value of property. M is an original property of the data value, M_{norm} is a normalized value of the data as represented by equation (1). Normalization enhances convergence and removes bias on features that are larger in magnitude.

Linear Discriminant Analysis (LDA) of features

LDA removes temporal, geographic, and socioeconomic aspects, decreasing dimensionality by maximizing between-class variation while minimizing within-class variance. Unlike PCA, which is an unsupervised technique that reduces dimensionality based only on overall data variance, LDA is preferred because it is a supervised method that utilizes class labels to enhance class separability. Equation (2) uses a within-class scatter matrix to improve separability by uncovering hidden tourist intensity and environmental correlations, which improves classification performance.

$$S_w = \sum_{i=1}^d \sum_{j=1}^{M_i} (w_j^i - \mu_i) (w_j^i - \mu_i)^S \quad (2)$$

$$S_b = \sum_{i=1}^d (\mu_i - \mu) (\mu_i - \mu)^S \quad (3)$$

In equation (3), $\sum_{j=1}^{M_i}$ represents the summation of all samples within a class. w_j^i is the sample feature vector in class, and μ_i is the mean feature vector of class. S_w is the minimize (make classes compact) and S_b is the maximize (increase class separation). S measures the spread of data points; it includes within-class and between-class scatter. w represents the variability of samples inside each class. b measures the separation between class means. i and j are the indices of class and sample. μ is the mean value. M is the number of samples in a class. d denotes the total number of classes. Identifies connections between tourism activities and environmental deterioration, including effects on vegetation loss, land-use change, and resource consumption, hence enhancing forecasting accuracy and interpretability.

Prediction model using KWA-Weighted RF for tourist demand and environmental impact assessment

A hybrid KWA-WRF model is proposed for predicting the tourist demand and environmental impact assessment. The WRF assign weights to attributes based on their contribution to tourism demand forecasting and environmental impact prediction. The KWA, a metaheuristic optimization strategy based on cooperative hunting behavior, was used to optimize the hyperparameters of the WRF model. This optimization improved classification accuracy by lowering overfitting and enhancing generalization performance.

Weighted RF improves reliability by focusing on key features and imbalance

The Weighted RF features forecasts tourism demand and environmental effects by emphasizing important features, addressing class imbalance, and merging multiple decision trees using bootstrap sampling and majority voting to improve accuracy and generalization. Out-of-bag samples aid internal review. Figure 2 show the weighted RF with OOB forecasts.

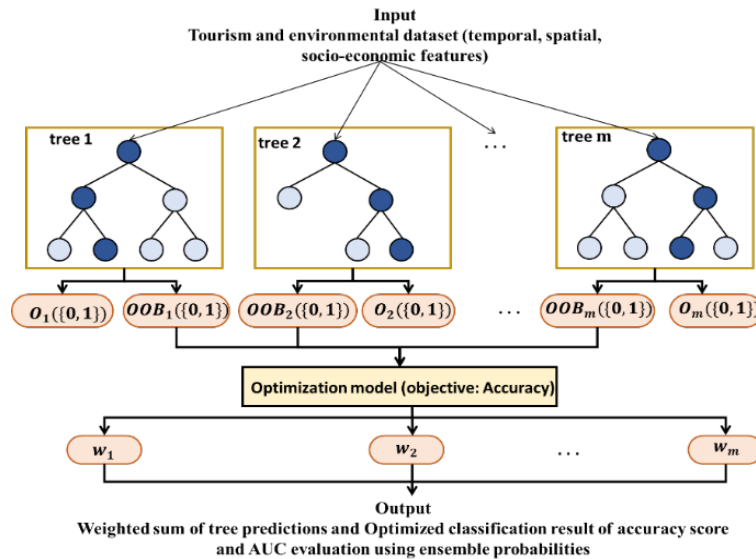


Figure 2. Weighted Random Forest Using Out-of-Bag Predictions

In classifier g_i , designate the collection of out-of-bag instances as P_i . $R(w, z_i)$ represents the out-of-bag proportion of values for class z_i at input w , and $O(g(w)=z_i)$ is an estimate. $\sum_{j=1}^m$ is the summation symbol starting index and ending index.

$$R(w, z_i) = \frac{\sum_{l=1}^L J(g_l(w)=z_i; (w, z) \in P_l)}{\sum_{l=1}^L J(g_l(w); (w, z) \in P_l)} \quad (4)$$

Equation (4), R is the out-of-bag (OOB) estimated class proportion or probability, and L is the total number of classifiers (trees) in the ensemble. J is the indication, and P represents the bootstrap sample or training subset utilized by each classifier. w is the input instance (feature vector) being classed, whereas z is the class label or target variable. i and l are the indices of a given class and classifier/tree in the ensemble. The classifier function (prediction model) is denoted by g , whereas $\in$ represents a set member. The metric function is represented by $J(g_l(w); (w, z) \in P_l)$. The range function measures the variance among the mean scores in class z and other classes, and is given by equation (5).

$$nq(w, z) = O(g(w) = z) - \max_{\substack{i=1 \\ i \neq z}} O(g(w) = z_i) \quad (5)$$

Here, n_q is the stronger more confident classification, $O(g(w)=z)$ denotes the output (or vote) supporting the correct class, while $\max_{i=1, i \neq z} O(g(w)=z_i)$ identifies the maximum output among all incorrect classes. n denotes the range function value, and q represents the class-wise scoring index by comparing prediction probabilities. Here, O denotes the out-of-bag probability estimate, and g is the classifier function that generates predictions for input w . The i refers to alternate class indices, and $\max$ finds the highest probability among all classes except z . It's approximated using $R(w, z)$ and $R(w, z_i)$. $\sum_{j=1}^m$ is the average over samples, and $R(w_j, z)$ is the correct-class score. Equation (6) is defined as the predicted margin, which is computed as the average throughout the training set.

$$T = \frac{1}{m} \sum_{j=1}^m \left(R(w_j, z) - \max_{\substack{i=1 \\ i \neq z}} O(w_j, z_i) \right) \quad (6)$$

The mean correlation is calculated as the difference of the edge times a square of a forest's variance, as shown by equation (7).

$$\bar{\rho} = \frac{\frac{1}{m} \sum_{j=1}^m \left(R(w_j, z) - \max_{\substack{i=1 \\ i \neq z}} O(w_j, z_i) \right)^2 - T^2}{\frac{1}{m} \sum_{l=1}^L \sqrt{(\rho_l + \hat{\rho}_l + (\rho_l - \hat{\rho}_l))^2}} \quad (7)$$

In equation (7), ρ denotes the mean correlation of the weighted random forest, where z is the actual tourism demand class label, m and L are the numbers of samples and trees, while j and l denote sample and tree indices. $R(w_j, z)$ and $O(w_j, z_j)$ represent correct and competing class probabilities, T is the prediction margin, Σ is the aggregation operator, and ρ_l , ρ_l are tree-wise correlation measures.

$$\rho_l = \frac{\sum_{(w_j, z \in P_l) J(g_l(w)=z)} J(g_l(w)=z)}{\sum_{(w_j, z \in P_l) J(g_l(w)=z)} J(g_l(w)=z)} \quad (8)$$

$$\hat{\rho}_l = \frac{\sum_{(w_j, z \in P_l) J(g_l(w)=\hat{z})} J(g_l(w)=\hat{z})}{\sum_{(w_j, z \in P_l) J(g_l(w)=\hat{z})} J(g_l(w)=\hat{z})} \quad (9)$$

Here, ρ^- and ρ^+ denote the correct-class and competing-class correlation scores of the l^{th} decision tree. P_l represents out-of-bag samples, $g_l(w)$ is the tree prediction for input w , z and $\hat{z}$ denote true and competing class labels, $J(\cdot)$ is the indicator function, and l indicates the tree index in the weighted random forest ensemble.

$$\hat{z}_i = \arg \max_{i=1} R(w, z_i) \quad (10)$$

Here, z_i denotes the highest probable incorrect class, while $\arg \max$ selects the class with maximum probability. i is the class index, z is the true class label, and $R(w, z_i)$ represents the out-of-bag probability for input feature vector w and candidate class z_i in the Weighted RF model. Optimal weighted random forest reduces prediction error by employing weights as choice factors and using out-of-bag cross-validation predictions to mimic unknown data for optimization.

$$\max \text{accuracy}(Z, [\sum_{j=1}^L x_i \hat{z}_i + 0.5]) \quad (11)$$

In equation (11), Z is the vector consisting of the actual reaction values, $\hat{z}_i$ is the out-of-bag forecast for a decision tree i , and x_i are weights related to a decision tree i . Equation (12) shows the:

$$\text{accuracy} \left(Z, \left[\sum_{j=1}^L x_i \hat{z}_i + 0.5 \right] \right)$$

Is the function that measures the ratio of correct forecasts (true positives and true negatives) over all analyzed cases, especially for unbalanced datasets.

$$\max \text{AUC} (Z, [\sum_{j=1}^L x_i \hat{o}_i]) \quad (12)$$

The out-of-bag probabilities vectors of all base classifiers are commonly referred to as in the preceding formulation, and the area under the ROC curve for the constructed ensemble is determined by:

$$\max \text{AUC} \left(Z, \left[\sum_{j=1}^L x_i \hat{o}_i \right] \right)$$

The suggested weighted RF enhances classification accuracy and resilience by optimizing tree weights with out-of-bag estimations, improving AUC performance, and successfully dealing with class imbalance in sustainable tourism prediction tasks.

KWA-Based Hyperparameter Optimization for Improved Performance

KWA optimizes model hyperparameters to enhance search efficiency, improve convergence speed, and increase prediction accuracy by identifying optimal parameter combinations for better model performance and generalization. Killer whales (*Orcinus orca*)⁽²¹⁾ are top predators in the marine ecosystem and play an ecological function and environmental influence in marine food webs.

The KWA flowchart sets up settings for clusters, matriline, and iterations, driving leader-based optimization.

It employs clustering to prevent local optima while improving global search efficiency. This approach contributes to tourism demand and environmental impact based on big data by increasing prediction accuracy and discovering optimal environmental-tourism balance locations, as outlined in the overall process shown in figure 3.

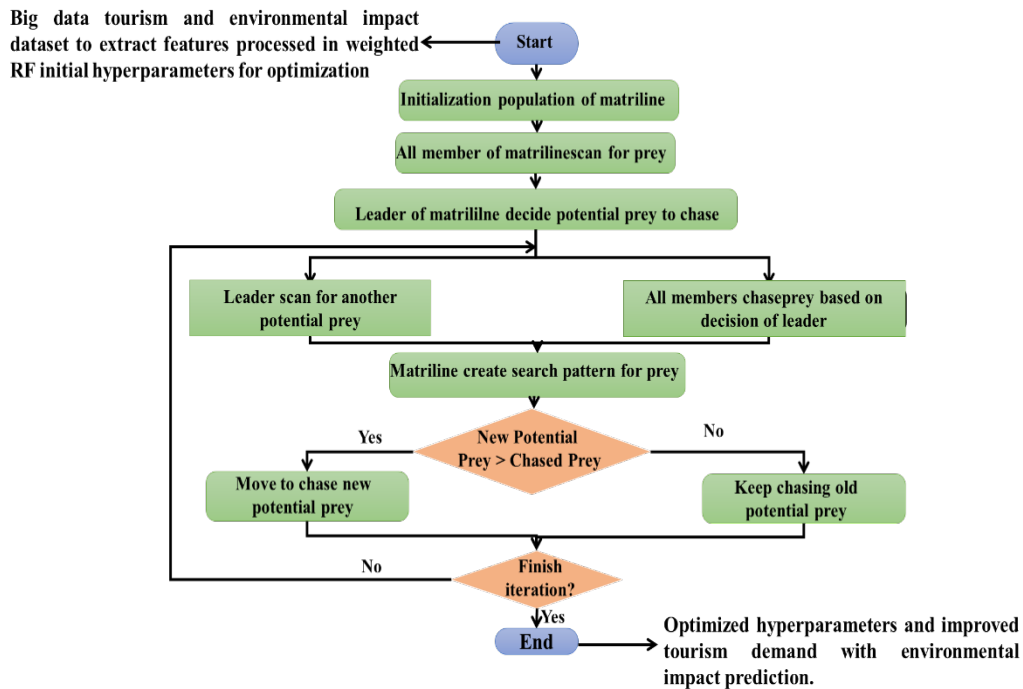


Figure 3. Flowchart of the KWA of the optimal environmental tourism

Killer whale echolocation is modelled using moving agents to replicate prey-seeking behavior. This mathematical model aids in determining the best solution of the goal function using angle θ , as specified in equation (13).

$$\theta = \sin^{-1} \left(\frac{c_E - c_p}{Q} \right) = \tan^{-1} \left(\frac{c_E - c_p}{W} \right) \quad (13)$$

The initial calculates the angle θ using the inverse sine ($\sin^{-1}$) and inverted tangent ($\tan^{-1}$) functions, depending on the difference between ecological constants c_E is the prey-related parameter c_p , Q and W is the normalized by factors signifying directional effect in the search space. Each search agent goes toward prey at a predetermined pace and direction, statistically modelling killer whale movement to update locations, and achieving the best results can be mathematically represented by the following equation (14):

$$\begin{cases} \vec{u}_k \leftarrow \vec{u}_k + \vec{V}(0, \phi_1) \oplus (\vec{o}_k - \vec{w}_k) + \vec{V}(0, \phi_2) \oplus (\vec{o}_h - \vec{w}_k) \\ \vec{w}_k \leftarrow \vec{w}_k + \vec{u}_k \cdot s \end{cases} \quad (14)$$

Here, u_k represents agent velocity, position (w_k), and best positions (o_k, o_h). u represents the velocity vector of the search agent, k is the agent index, w denotes the position vector, and s is the step size. o indicates target-related position vectors, h is another guiding agent, $\vec{V}(0, \phi_1)$ is a random velocity influence, and $\oplus$ represents a stochastic or element-wise combination operator. KWA divides the search space into clusters, and agents explore inside them to discover optimum solutions. Each agent starts from a cluster center and works together to find the best local answer. KWA improves optimization efficiency, convergence, and properly identifies optimal solutions, hence facilitating sustainable tourism modelling and environmental impact assessment using large data.

RESULTS

The experiment using Python-based simulations employs performance measurements to determine the optimal ability and efficiency of feature-based big data used for tourism demand and environmental impact. The dataset is separated into three categories: 70 % training, 15 % testing, and 15 % validation. Table 1 summarizes

key hyperparameters, preprocessing settings, and optimization ranges used in the proposed KWA-Weighted Random Forest model for reproducible tourism demand and environmental impact prediction experiments. Table 1 hyperparameter Settings and Optimization Configuration of the Proposed KWA-Weighted Random Forest Model.

Table 1. Hyperparameter and Optimization Configuration of the Proposed KWA-Weighted RF		
Component	Hyperparameter	Value
Data Preprocessing	Missing value handling	Mean/Interpolation
	Normalization	Min-Max (0-1)
Feature Extraction (LDA)	Method	Linear Discriminant Analysis
	Objective	Maximize class separability
Weighted Random Forest	Number of trees	100-300
	Max depth	10-30
	Min samples split	2-6
	Min samples leaf	1-4
	Criterion	Gini
	Sampling	Bootstrap + OOB enabled
KWA Optimization	Population size	30
	Iterations	100
	Objective	Accuracy + AUC maximization
Training Setup	Split ratio	80:20
	Validation	10-fold CV
	Seed	42
Environment	Platform	Python
Hardware Configuration	Processor (CPU)	Intel Core i7
	GPU	NVIDIA GPU
	RAM	16 GB

Performance evaluation of the sustainable tourism and environment impact assessment

Training and validation accuracy continuously improve as loss decreases, showing strong learning, steady convergence, and low overfitting across all epochs. Figure 4(a) displays training and validation performance increases slowly, with about 0,70 as training and 0,68 as validation. Figure 4(b) shows a decrease in loss between approximately 1,10 and 0,10 and 1,15 and 0,15, respectively. The presence of a small gap implies successful generalization and minimal overfitting during the examination of the learning process.

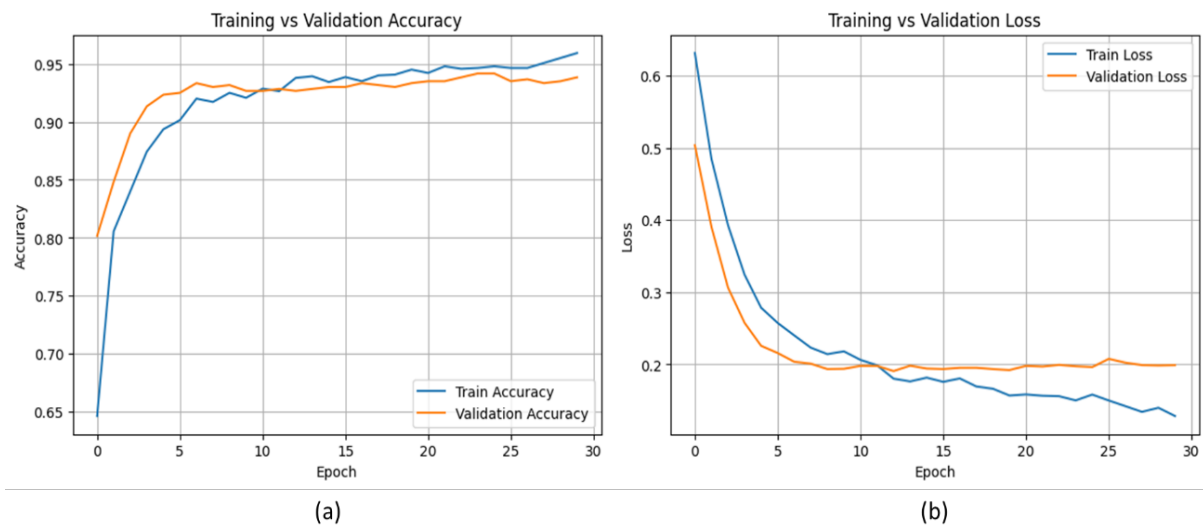


Figure 4. Evaluation of the model performance in terms of training/validation (a) accuracy and (b) loss of sustainable tourism

The confusion matrix shows accurate predictions on the diagonal, indicating that most samples were successfully classified into three groups with small errors in misclassifications. The results of the three-class model assessment, the confusion matrix shows the correct predictions 3009, 3007, and 2933 and incorrect predictions 71,55,48,68, 57, and 73. Additionally, error analysis shows that misclassifications occur between Class-1 and Class-2 (71 and 55 cases), Class-2 and Class-3 (48 and 68 cases), and Class-1 and Class-3 (57 and 73 cases), indicating that closely related feature distributions lead to these overlaps, as reflected in figure 5.



Figure 5. Confusion matrix indicating strong classification performance

The visitors’ flows (28 %), mobility patterns (22 %), lodging use (20 %), seasonal changes (18 %), and transit demand (12 %) are major determinants of tourism dynamics, which implies their relative importance in tourism planning and management. Figure 6 uses visitor flow to represent tourist numbers, seasonal variation to describe demand variation, mobility patterns to describe movement behavior, accommodation use to describe lodging demands, and transit demand to assess infrastructure needs to conduct effective tourism operations.

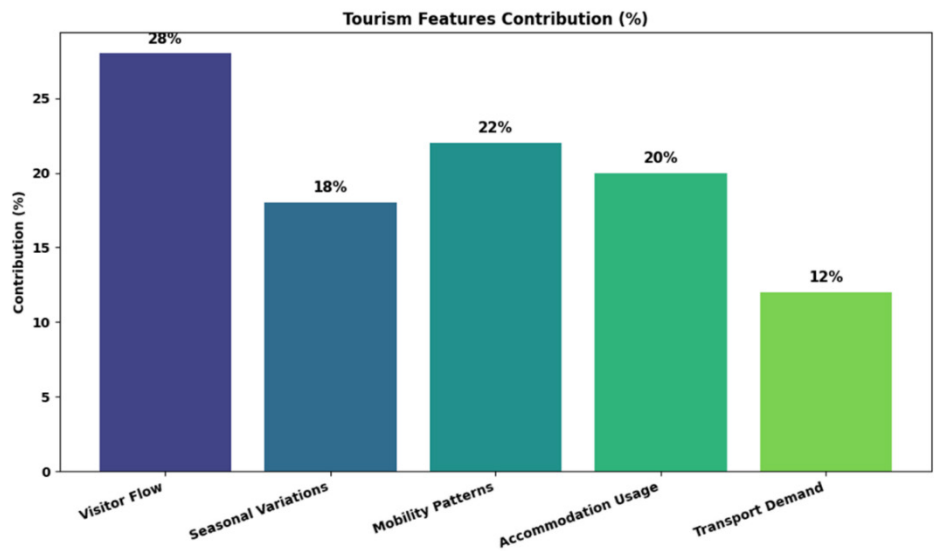


Figure 6. Contribution Analysis of Tourism Features

The model has high training and validation accuracy, low loss, and a confusion matrix, feature contribution to tourism, with generally correct classifications with minimal misclassifications, and good generalization ability.

Comparative analysis

The results reveal that the suggested predictive model greatly improves sustainable tourist demand and environmental impact assessment compared to baseline strategies. eXtreme Gradient Boost (XGBoost),⁽¹⁹⁾ Deep Learning Neural Network (DLNN),⁽¹⁹⁾ Support Vector Machine with Red Deer Optimization (SVM-RDO),⁽²⁰⁾ and

KWA-Weighted RF based on big data.

Accuracy assesses total forecast accuracy, which supports accurately monitoring tourists' environmental effect trends. Precision rates correctly anticipated positives, guaranteeing that the detection of environmental deterioration is accurate. Recall measures the capacity to detect true positives, which is consistent with the impact monitoring. F1-Score is the balance between accuracy and recall, and therefore it fulfills the objective of examination, which is to produce accurate, sustainable tourism effect evaluation. Table 2 compared between the conventional model and KWA-weighted RF showed that the values of accuracy, precision, recall, and F1-score were 98,7 %, 96,3 %, 95,1 %, and 95,7 %, respectively, beating the SVM-RDO model, whose respective values were 89,2 %, 90,2 %, and 89,8 %.

Methods	Dataset	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)
XGBoost	Multi-source spatial datasets	82,4	-	-	-
DLNN		84,3	-	-	-
SVM-RDO	Tourism dataset	92	89,2	90,2	89,8
KWA-Weighted RF [Proposed]	Tourism and environment impact dataset	98,7	96,3	95,1	95,7

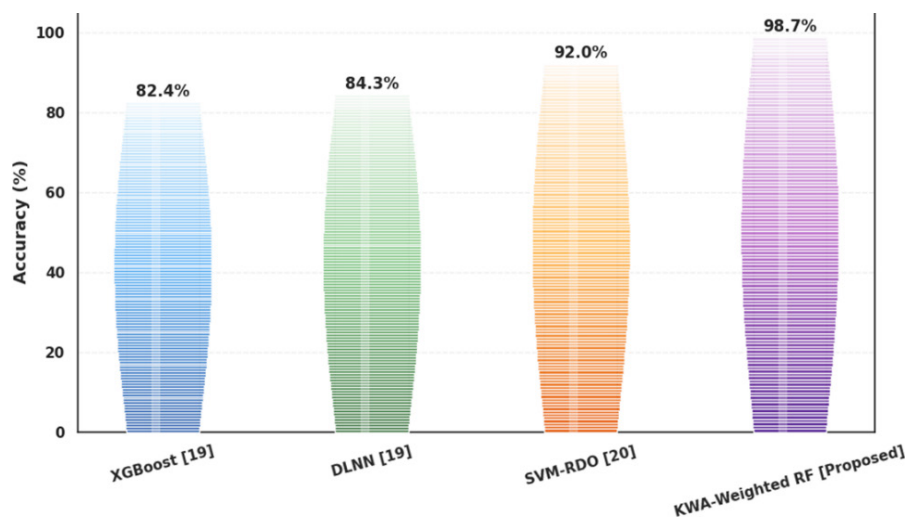


Figure 7. Comparison of the Accuracy of KWA-Weighted RF and existing models

The suggested KWA-weighted RF outperforms previous models. As can be seen in figure 7, the accuracy is 98,7 %, better than the SVM-RDO method (92 %), the DLNN method (84,3 %), and the XGBoost method (82,4 %).

Kappa is a statistical measure designed to assess the reliability of classification when predicting the tourist impact model. The average absolute prediction error, as estimated by MAE, assists in attaining the objective of minimizing the deviations in the environmental impact assessments. RMSE is a squared error scale that prioritizes higher deviations, enhancing tourism demand and the accuracy of environmental predictions. AUC to determine high-impact cases and low-impact cases of a tourist scenario. Table 3 is a comparison of the conventional model to the KWA-weighted RF.

Methods	Dataset	Kappa	MAE	RMSE	AUC
XGBoost	Multi-source spatial datasets	0,732	0,216	0,345	0,905
DLNN		0,781	0,197	0,283	0,914
KWA-Weighted RF [Proposed]	Tourism and environment impact dataset	0,964	0,112	0,168	0,982

The suggested KWA-Weighted RF has improved performance, with a Kappa of 0,964, MAE of 0,112, RMSE of 0,168, and AUC of 0,982. Figure 8 shows that the results beat DLNN (0,781, 0,197, 0,283, 0,914) and XGBoost (0,732, 0,216, 0,345, 0,905), showing greater dependability and lower prediction error.

Table 4 provides existing models, including XGBoost, DLNN, and SVM-RDO, that were trained for tourism

demand and environmental impact prediction. The results show that the KWA-Weighted RF performed excellently with an accuracy of 98,7 %, precision of 96,3 %, recall of 95,1 %, F1-score of 95,7 %.

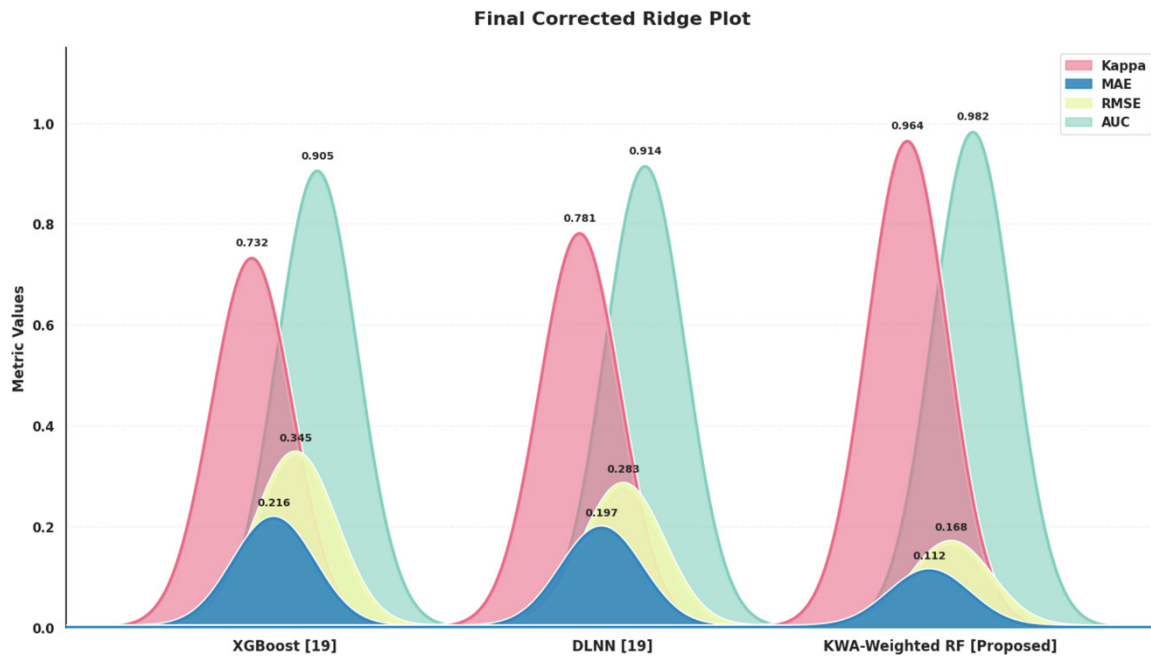


Figure 8. Comparison of the KWA-Weighted RF and existing model

Table 4. Tourism and environment impact Dataset to train the standard model and KWA-Weighted RF

Methods	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)	AUC
XGBoost	83,6	81,4	82,1	81,7	0,912
DLNN	85,9	84,2	83,6	83,9	0,921
SVM-RDO	93,2	90,6	91,3	90,9	0,946
KWA-Weighted RF [Proposed]	98,7	96,3	95,1	95,7	0,982

Overall, KWA-Weighted RF beats XGBoost, DLNN, and SVM-RDO in terms of accuracy, precision, recall, F1-score, Kappa, MAE, RMSE, and AUC, indicating higher reliability for tourism demand prediction and environmental impact assessment.

5-Fold Cross validation

Cross-validation is used to evaluate model robustness by splitting data into five subsets and testing iteratively. It provides reliable performance estimation by reducing bias and variance. In this study, it ensures stable evaluation of the proposed KWA-Weighted RF model.

Table 5. 5-Fold Cross-Validation Results of the Proposed KWA-Weighted RF Model

Fold	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)
1	98,4	95,9	94,8	95,2
2	98,6	96,2	95,0	95,5
3	99,0	96,8	95,6	96,0
4	98,5	96,1	95,1	95,6
5	99,1	96,5	95,2	96,0
Mean ± SD	98,72 ± 0,29	96,30 ± 0,31	95,14 ± 0,29	95,66 ± 0,27
95 % CI	98,36, 99,08	96,02, 96,58	94,88, 95,40	95,41, 95,91
p-value (t-test)	< 0,001	< 0,001	< 0,001	< 0,001

Table 5 presents 5-fold cross-validation results of the proposed KWA-Weighted Random Forest model for tourism demand and environmental impact prediction. Each fold shows consistent performance across metrics.

The mean $\pm$ standard deviation, while statistical significance is confirmed using t-test ($p < 0,001$), indicating stable, reliable, and robust model performance across different data splits.

Ablation Study

The ablation study evaluates the contribution of each component in the proposed framework, including normalization, LDA based feature extraction, KWA optimization, and weighted learning. Results show that each component individually improves performance over the baseline RF, while the full integrated model achieves the highest performance across all evaluation metrics shown in table 6, confirming the effectiveness of the complete pipeline.

Model Variant	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)	AUC
RF (Baseline)	93,8	90,2	91,0	90,6	0,948
RF + Min-Max Normalization	95,2	91,8	92,3	92,0	0,956
RF + LDA Features	95,7	92,9	92,1	92,4	0,961
RF + KWA (without optimization)	96,1	93,4	92,8	93,0	0,964
RF + Min-Max + LDA + KWA + Weighted Learning (Full Model)	98,7	96,3	95,1	95,7	0,982

DISCUSSION

The big data based on the tourist demand and environmental impact assessment, also addresses the main issues of assessing the research's past outcomes. Relies on a small number of contextual datasets, lacks scalability, complicates ontology maintenance, and provides insufficient assessment across varied locations and changing tourist contexts.⁽¹⁸⁾ It is reliant on regional data, with low generalization, high to compute, and limited integrated with other systems, including deep learning, machine learning, and decision analysis.⁽¹⁹⁾ The shortcomings of optimized SVM include parameter optimization, handling of large heterogeneous datasets, and difficulty in interpretation when applied to complex tourists' behavior pattern applications.⁽²⁰⁾ KWA-Weighted RF Overcomes this limitation through the inclusion of heterogeneous big data, transformation of parameters, scalability, decrease in their computational complexity, interpretability, and improved forecasting and prediction of tourism demand and environmental effects.

Practical implications: big data in tourism include improved policy-making, real-time environmental monitoring, efficient resource allocation, and reduced environmental impact, supporting sustainable tourism planning. The predicted high-impact zones assist policymakers in regulating tourist flow, enhancing infrastructure, and protecting sensitive areas. The model also offers interpretability by identifying key contributing factors such as tourist density, seasonal variation, and environmental stress. Additionally, the framework shows potential for generalization to other regions, though performance may vary in data-scarce settings and may require adaptation or transfer learning.

CONCLUSIONS

Sustainable tourism and environmental impact assessment are effectively supported using KWA-Weighted Random Forest with big data analytics. The framework applies preprocessing and LDA-based feature extraction to improve data quality and identify meaningful relationships among variables.

The optimized model enhances prediction of tourism demand and environmental impacts while supporting reliable and balanced decision-making.

The model shows strong performance with accuracy, indicating robust and reliable prediction capability. Overall, the results demonstrate strong predictive capability and practical usefulness for surveillance, tourism planning, and environmental management without relying on repeated statistical details. This work enables governments and tourism boards to make data-driven decisions that balance economic growth with ecosystem protection, reduce environmental degradation, and support long-term sustainable tourism planning at regional and national levels. Limitations include data heterogeneity, computational complexity, data dependency, and limited interpretability of environmental factors.

Future work includes (i) integrating IoT sensors for real-time monitoring, (ii) extending datasets across regions and seasons, (iii) applying XAI methods (SHAP/LIME) for interpretability, and (iv) developing a decision-support dashboard for real-time policy insights. Future work includes integrating real-time IoT sensors for continuous environmental monitoring, expanding datasets across multiple regions and seasons for better generalization, applying explainable AI for transparent policymaking, and deploying a decision-support dashboard for tourism authorities. **Broader impact:** Enables data-driven, sustainable tourism governance that balances economic

growth with long-term environmental conservation and climate resilience.

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CONFLICT OF INTEREST

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